

GMD (Software for mobile device)

Recycler view & cardview

A recycler view is alternative & modified version of list view - It loads the data into memory in the form of chunks -

Add dependency first → 'androidx.recyclerview:
'gradle script' recyclerview:1.2.0'
'androidx.cardview:
cardview:1.0.0'

→ first make a layout in layout file

→ then make a personAdapter in the java -

If we have to make ~~person~~ Adapter
ourselves we have our own style of data -
we cannot make builtin adapter of person -

```
public class PersonAdapter <PersonAdapter.ViewHolder>  
    RecyclerView.Adapter {  
        ArrayList<Person> people;
```

```
    public PersonAdapter(Context context, ArrayList  
        <Person> list)  
    {  
        people = list
```

@Override
 public int getItemCount ()

// we also need to hold our view
so we need one more class view holder
class

```
public class ViewHolder extends  
RecyclerView.ViewHolder
```

code of RecyclerView -

// create a model class first - which you
want to display your information

```
public class Person  
{
```

```
    String name, surname, preference  
}
```

// want to make our design -
res → layout

// now we will make our custom
Adapter because we have our own
data here -

```
public class PersonAdapter extends RecyclerView
```

```
Adapter < PersonAdapter.ViewHolder >
```

```
{    ItemSelected parentActivity;
```

```
    ArrayList < Person > People;
```

```
    public interface ItemSelected
```

```
{
```

```
        public void onItemClick (int index);
```

```
}
```


Recycler view shows the records dynamically - It is used to scrolls ~~horizontally~~ ^{vertically} not ~~a~~ Horizontal list view can scroll ~~vertically~~ ^{vertically or horizontally} too

```
public PersonAdapter(Context context, ArrayList<Person> list)
```

```
{
```

```
    ParentActivity = context;
```

```
    people = list;
```

```
}
```

```
public class ViewHolder extends RecyclerView.ViewHolder
```

```
{
```

```
    ImageView ipref;
```

```
    TextView tname, sname
```

```
    public ViewHolder(@NonNull final  
        View itemView) {
```

```
        super(itemView);
```

```
        ipref = itemView.findViewById(R.id.ipref);
```

```
        tname = itemView.findViewById(R.id.tname);
```

```
        sname = itemView.findViewById(R.id.sname);
```

```
        itemView.setOnClickListener(new View.OnClickListener() {
```

```
            @Override
```

```
            public void onClick(View v)
```

```
            {
```

```
                ParentActivity.onItemClick(People.indexOf(Person  
                    itemView.getTag());
```

```
            })  
        }  
    }
```


@NotNull
@Override

public ViewHolder onCreateViewHolder(@NotNull View
parent, int viewType)

{

View v = LayoutInflater.from(parent.getContext())
inflate(R.layout.list_item, parent, false)
return new ViewHolder(v);

}

@Override

public void onBindViewHolder(@NotNull ViewHolder
holder, int position)

{

holder.itemView.setTag(people.get(position))

holder.tvname.setText(people.get(position)
getName());

holder.tvname.setText(people.get(position)
getName());

if (people.get(position).getPreference().equals("bus")

{

^{imageView}
holder.setImageResource(R.drawable.bus);

}

else if (people.get(position).getPreference().equals
("car"))

{

holder.imageView.setImageResource(R.drawable.car);

}

~~@Override~~

public class MainActivity extends AppCompatActivity
implements PersonAdapter.ItemSelected {

RecyclerView recyclerView;

RecyclerView.LayoutManager manager;

RecyclerView.Adapter myAdapter;

ArrayList<Person> person;

Button AddButton;

@Override

protected void onCreate(Bundle savedInstanceState)
{

super.onCreate(savedInstanceState);

setContentView(R.layout.activity_main);

~~btn~~ ^{Button} findViewById(R.id.AddButton);

RecyclerView findViewById(R.id.~~list~~ list);

recyclerView.setHasFixedSize(true);

manager = new LinearLayoutManager(this,
2, LinearLayoutManager.HORIZONTAL, false);

recyclerView.setLayoutManager(manager);

person = new ArrayList<>();

person.add(new Person("Data"));

person.add(new Person("Data"));


```
btnButton.setOnClickListener (new View.OnClickListener () {
```

```
public void onClick (View v)
```

```
{
```

```
new
```

```
Person.add (Person ("Ab"));
```

```
myAdapter.notifyDataSetChanged();
```

```
}
```

```
}  
@Override
```

```
myAdapter = new PersonAdapter (this, Person);
```

```
recyclerView.setAdapter (myAdapter);
```

```
@Override
```

```
public void onItemClick (int index)
```

```
{
```

```
Toast.makeText (this, person.get(index).
```

```
getName (), Toast.
```

```
LENGTH_SHORT).show();
```

```
}  
}
```


shared preferences = name as in the
webdevelopment

ViewPager → gives us option to swap
the fragments

^{set}
~~give~~ up three fragments (chats, status and
calls) -

→ Main Activity.java =

TabLayout tabLayout; // for tab layout

ViewPager2 viewPager2; // for swap

TabLayoutMediator mediator; // connect tabLayout &
viewpager
boolean isFirst;

SharedPreferences spref; // shared preference
internal file to store login

init()

private void init();

{ tabLayout = findViewById(R.id.tabLayout);

viewPager = findViewById(R.id.viewPager2);

adapter = new ViewPager2Adapter(this);

isFirst = false

name:

spref = getSharedPreferences("user", Private mode,

viewPager2.setAdapter(adapter);

mediator = new TabLayoutMediator(tabLayout,

viewPager, () {

this, (ONCLICK, SHORT).show()


```

@Override
public void onConfigureTab (@NotNull TabLayout, int tab position) {
    switch (position) {
        case (0):
            tab.setText("Chats") // Attach name of tab
            tab.setIcon(R.drawable.chats_icon)
            BadgeDrawable bd = tab.getOrCreateBadge()
            bd.setNumber(100);
            bd.maxCharacterCount(2);
            break;
        case (1):
            tab.setText("Status")
            tab.setIcon(R.drawable.updates_icons);
            break;
        case (2):
            tab.setText("Calls");
            tab.setIcon(R.drawable.icons_calls);
            BadgeDrawable bd = tab.getOrCreateBadge();
            bd.setNumber(8)
            bd.setMaxCharacterCount(3)
            break;
    }
}

mediator.attach();

```



```
ViewPager2.registerOnPageChangeCallback(new ViewPager2.  
OnPageChangeCallback() {
```

```
    @Override
```

```
    public void onPageSelected (int position)  
    {
```

```
        if (isFirst)  
        {
```

```
            BadgeDrawable bd = tabLayout.getTabAt(position)
```

```
            .getOrCreateBadge();
```

```
            bd.setNumber(0);
```

```
            bd.setVisibility(false);  
        }
```

```
        else
```

```
        {
```

```
            isFirst = true
```

```
        }  
    }  
}
```

```
public void checkedAlreadyLogin(); // security gatekeeper -
```

```
    { // it decides whether
```

```
        if (!spref.getBoolean("isLogin", false)) // the user will go to  
        { // MainActivity screen
```

```
        { // on login screen -
```

```
            startActivity(new Intent(this, LoginActivity.class));
```

```
        }  
    }  
}
```



```
public class ViewPagerAdapter extends FragmentState  
Adapter {
```

```
public ViewPagerAdapter (FragmentActivity  
fragmentActivity) {
```

```
super (fragmentActivity);
```

```
@NotNull
```

```
@Override
```

```
public void createFragment (int position)
```

```
{  
    switch (position)
```

```
    {  
        case 0:
```

```
            return new Chat();
```

```
        case 1;
```

```
            return new Updater();
```

```
        case 2;
```

```
    }  
}
```